When Glashow, Weinberg, and Salam were developing these ideas, the W and Z particles had yet to be discovered experimentally. It was the strong faith these physicists had in the power of theory and the beauty of symmetry that gave them the confidence to go forward. Their boldness proved well founded. In due course, the W and Z particles were discovered and the electroweak theory was confirmed experimentally. Glashow, Weinberg, and Salam had looked beyond superficial appearances—they had peered through the obscuring fog of nothingness—to reveal a deep and subtle symmetry entwining two of nature's four forces. They were awarded the 1979 Nobel Prize for the successful unification of the weak nuclear force and electromagnetism.

Grand Unification

When I was a freshman in college, I'd drop in every now and then on my adviser, the physicist Howard Georgi. I never had much to say, but it hardly mattered. There was always something that Georgi was excited to share with interested students. On one occasion in particular, Georgi was especially worked up and he spoke rapid fire for over an hour, filling the chalkboard a number of times over with symbols and equations. Throughout, I nodded enthusiastically. But frankly, I hardly understood a word. Years later I realized that Georgi had been telling me about plans to test a discovery he had made called *grand unification.*

Grand unification addresses a question that naturally follows the success of the electroweak unification: If two forces of nature were part of a unified whole in the early universe, might it be the case that, at even higher temperatures, at even earlier times in the history of the universe, the distinctions among three or possibly all four forces might similarly evaporate, yielding even greater symmetry? This raises the intriguing possibility that there might actually be a single fundamental force of nature that, through a series of cosmological phase transitions, has crystallized into the four seemingly different forces of which we are currently aware. In 1974, Georgi and Glashow put forward the first theory to go partway toward this goal of total unity. Their *grand unified theory,* together with later insights of Georgi, Helen Quinn, and Weinberg, suggested that three of the four forces—the strong, weak, and electromagnetic forces— were all part of one unified force when the